Giuseppe Cosimo Alfio **Bellamacina**

AI/ML Engineer for LLM agents, RAG and RL fine-tuning, 3+ years in production

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Summary

AI/ML Engineer with **3+ years** shipping LLM systems to production in regulated sectors: **public healthcare** and **military** (Italian Air Force Academy). Focused on multi-agent architectures, RAG at scale and model adaptation (fine-tuning, GRPO/RL, distillation). M.Sc. student in Machine Learning & AI.

Experience

AI Software Engineer at **RICCA IT** — Sep 2025 – Present

- Architected and deployed a **multi-agent system (∼10 agents)** serving **public healthcare organizations across southern Italy** (LangChain, LangGraph)
- Built RAG pipelines over **4M+ records** with **sub-second retrieval** (LLM-generated tag embeddings, vector search)
- Shipped AI modules as REST APIs (FastAPI), containerized with Docker, CI/CD on GitHub Actions

AI/ML Engineer at **IntelliSync** — Jun 2023 – Jul 2025

- Deployed RAG for the **Italian Air Force Academy**, a security-sensitive context serving **∼1,000 cadets/year**
- Built agentic text-to-query pipelines over **SQL, Cypher, SPARQL and vector search** (LangGraph)
- Delivered anomaly detection and energy forecasting for **30+ wind farms** across Europe and MENA (Yokogawa)

Projects

Guardian **Hackathon winner, NeoData Hackatania 2.0 (2024)**
Multimodal AI assistant for law enforcement: **4-agent system** with **GraphRAG over a 28M-node Neo4j graph**; correlates identities from image inputs (e.g. license plates) and assesses risk in real time (LangChain, LangGraph).

Neuro-Symbolic Text-to-Gloss **Research, neuro-symbolic NLP**
English-to-ASL gloss with Qwen2.5-0.5B, **Trie-constrained decoding** and **SFT → GRPO**: **ROUGE-L 0.137 → 0.974** vs zero-shot, **81% exact match**, 100% valid output (Trie-enforced); 15-cell ablation study.

Warehouse Swarm Intelligence **Research, multi-agent systems**
Decentralized scout/coordinator/retriever swarm exploring unknown warehouses with limited communication: frontier exploration, A* replanning, atomic task claims. Real-time simulator (FastAPI, Socket.IO, React).

PitWall AI **Personal project**
Fully local AI F1 race engineer: real Grands Prix replayed through **Apache Kafka and Spark Structured Streaming** for race strategy; a local LLM (Ollama) commentates and answers via **tool calling**. Dockerized microservices.

Technical Skills

Languages	Python, C, C++, SQL, Cypher, SPARQL, Bash, JavaScript, TypeScript
LLM & Agents	LangChain, LangGraph, LangSmith, HuggingFace, vLLM, Ollama, tool calling, multi-agent systems
RAG & Search	GraphRAG, ChromaDB, Weaviate, FAISS, vector search, semantic search, reranking
Fine-tuning	SFT, GRPO/RL, LoRA/QLoRA, Unsloth, constrained decoding, distillation, domain adaptation
ML	PyTorch, TensorFlow, scikit-learn, Pandas, Weights & Biases, anomaly detection, forecasting
Data	Neo4j, PostgreSQL, MySQL, Redis, SQLite, Apache Kafka, Apache Spark
APIs & Web	FastAPI, Flask, REST API design, SDK integration, Socket.IO, React
DevOps & Cloud	Docker, GitHub Actions, CI/CD, Slurm (HPC), multi-GPU clusters (H100), AWS, Vercel, Render

Education

M.Sc. Machine Learning and Artificial Intelligence at Università degli Studi di Catania — 2025 – 2027 (expected)
B.Sc. Computer Science, 110/110 cum laude at Università degli Studi di Catania — 2021 – 2024

Certifications

LangChain: Deep Agents, Deep Research, Ambient Agents, Introduction to LangGraph, LangSmith (2025)
Anthropic: Claude Code in Action, MCP Advanced Topics, Introduction to MCP (2025)